Deepfake Image Detection Using Multi-Level Frequency Analysis

Comprehensive Architectural, Mathematical, and Empirical Research Dossier

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Primary Domain: Digital Image Forensics & Computer Vision
Core Innovation: Dual-Branch Spatio-Frequency Network with Wavelet Packet Attention
Evaluation Benchmark: 140k Real & Fake Faces (FFHQ vs. StyleGAN2) across c0, c23, c40
Key Metric: 99.52% Acc / 1.0000 AUC on c0; 98.51% Acc on aggressive c40 JPEG compression
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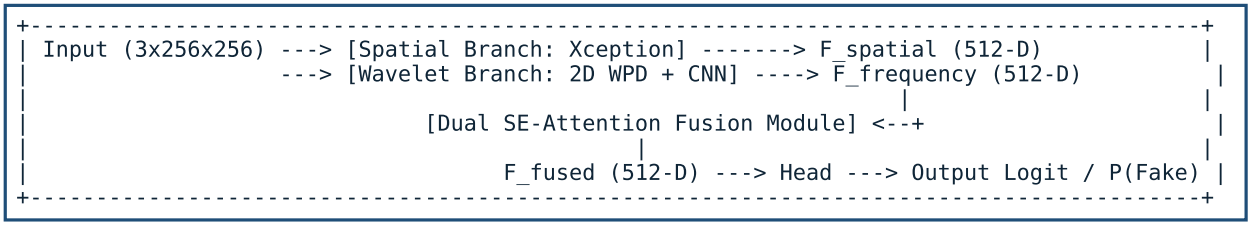
1.1 Problem Statement & Research Gap

Modern deep learning face forgery detectors achieve near-perfect discrimination in uncompressed laboratory settings. However, when deepfakes are distributed through social media networks (e.g., WhatsApp, X/Twitter, Instagram), lossy JPEG compression destroys high-frequency details, causing traditional spatial-only detectors to suffer performance drops of 10% to 15%. Global frequency transforms (such as standard DFT or DCT) fail because they lack spatial localization and cannot distinguish generative GAN artifacts from benign JPEG quantization noise.

1.2 Proposed Dual-Branch Spatio-Frequency Paradigm

This research introduces a Dual-Branch Spatio-Frequency architecture that explicitly decouples visual semantic analysis from directional spectral energy decomposition:

1. Spatial Context Branch: Leverages a pre-trained Xception CNN backbone to capture structural inconsistencies (irregular pupils, asymmetric facial landmarks, boundary blending artifacts).
2. Multi-Scale Wavelet Packet Branch: Deploys 2D Wavelet Packet Decomposition (WPD) using a Daubechies-4 (db4) basis up to Level 2. WPD recursively decomposes both low- and high-frequency octaves, isolating subtle transposed-convolution upsampling grid patterns across 16 sub-bands.
3. Cross-Domain SE-Attention Fusion: Squeeze-and-Excitation channel attention dynamically recalibrates domain reliance. Under heavy compression (c40), the module automatically down-weights corrupted frequency sub-bands and shifts attention to resilient spatial facial landmarks.



2.1 Spatial Context Branch Formulations

Let I in R^{B x 3 x 256 x 256} represent a normalized mini-batch of facial crops.
The spatial feature map from the deepest stage of Xception is denoted by X_spatial in R^{B x 2048 x 8 x 8}.
Global Average Pooling (GAP) collapses spatial dimensions across height and width:

GAP(X_spatial)_c = (1 / (H * W)) * sum_{h=1}^H sum_{w=1}^W X_spatial[:, c, h, w] in R^{B x 2048}

The spatial embedding vector F_spatial in R^{B x 512} is projected via a linear layer and batch norm:

F_spatial = ReLU(BatchNorm1D(W_proj_s * GAP(X_spatial)))

2.2 Multi-Scale 2D Wavelet Packet Decomposition (WPD)

Given scaling function $\phi(t)$ and mother wavelet $\psi(t)$, the 2D separable wavelet basis produces four fundamental sub-band filtering kernels at each recursive decomposition step:

psi^{LL}(x, y) = phi(x)*phi(y) [Approximation Octave]
psi^{LH}(x, y) = phi(x)*psi(y) [Horizontal Detail Octave]
psi^{HL}(x, y) = psi(x)*phi(y) [Vertical Detail Octave]
psi^{HH}(x, y) = psi(x)*psi(y) [Diagonal Detail Octave]

Unlike standard DWT which only decomposes the low-frequency approximation sub-band (LL), 2D WPD recursively decomposes all four quadrants up to depth L = 2, producing 4^L = 16 directional sub-bands:

W = { W_1, W_2, ..., W_16 }, where each sub-band W_i in R^{B x 64 x 64}

2.3 Log-Energy Sub-band Transform & CNN Stem

Because detail sub-band amplitudes exhibit high dynamic variance, we apply a log-energy transform:

W_tilde_i = log(|W_i| + epsilon), where epsilon = 1e-8

The 16 transformed sub-bands are stacked into tensor W_tilde in R^{B x 16 x 64 x 64} and processed by a lightweight 3-stage convolutional neural network stem:
* Stage 1: Conv2D(16 -> 64, k=3, s=2, p=1) -> BN -> ReLU ==> (B, 64, 32, 32)
* Stage 2: Conv2D(64 -> 128, k=3, s=2, p=1) -> BN -> ReLU ==> (B, 128, 16, 16)
* Stage 3: Conv2D(128 -> 256, k=3, s=2, p=1) -> BN -> ReLU ==> (B, 256, 8, 8)
* GAP(Stage 3) ==> (B, 256) -> Linear + BN + ReLU ==> F_frequency in R^{B x 512}

3.1 Cross-Domain Dual-Attention Fusion Mechanism

To adaptively modulate feature dominance based on image corruption, independent Squeeze-and-Excitation bottlenecks compute dynamic channel-wise gating vectors alpha_s, alpha_f in R^{B x 512}:

alpha_spatial = sigmoid(W_s2 * ReLU(W_s1 * F_spatial))
alpha_frequency = sigmoid(W_f2 * ReLU(W_f1 * F_frequency))

where reduction ratio r = 8, W_1 in R^{64 x 512}, W_2 in R^{512 x 64}, and sigmoid maps to (0, 1). Features are recalibrated via element-wise Hadamard product, concatenated, and projected:

F_hat_spatial = F_spatial * alpha_spatial, F_hat_freq = F_frequency * alpha_frequency
F_fused = Dropout_0.3(ReLU(BatchNorm1D(W_fuse * [F_hat_spatial || F_hat_freq]))) in R^{B x 512}

The binary classification head computes raw scalar logit y_hat:

y_hat = W_cls * Dropout_0.5(F_fused) + b_cls in R^{B x 1}, P(Fake) = sigmoid(y_hat)

3.2 Objective Function: Binary Cross-Entropy with Logits

For batch size B and binary ground-truth label y_i in {0, 1} (0 = Real, 1 = Fake):

L_BCE = - (1 / B) * sum_{i=1}^B [y_i * log(sigma(y_hat_i)) + (1 - y_i) * log(1 - sigma(y_hat_i))]

Numerically stable implementation evaluated via log-sum-exp formulation:

L_i = max(y_hat_i, 0) - y_hat_i * y_i + log(1 + exp(-|y_hat_i|))

3.3 Optimizer & Cosine Annealing Learning Rate Schedule

Model parameters are optimized via AdamW (LR = 1e-4, weight decay lambda = 1e-4, beta_1 = 0.9, beta_2 = 0.999). The learning rate follows a 2-epoch linear warmup followed by Cosine Annealing decay down to eta_min = 1e-6:

eta_t = eta_min + 0.5 * (eta_base - eta_min) * (1 + cos(pi * (t - 2) / 28)), for epoch t > 2

4.1 Benchmark Dataset Profile (140k Real & Fake Faces)

- Authentic Real Class: 70,000 real portraits from Flickr-Faces-HQ (FFHQ) aligned crops.
- Synthetic Fake Class: 70,000 photorealistic faces synthesized via StyleGAN / StyleGAN2.
- Train / Val / Test Partition: 100,000 train (50k/50k) | 20,000 val (10k/10k) | 20,000 test (10k/10k).
- On-the-fly JPEG Augmentation: Simulates quality factors $Q \in [23, 100]$ during training.

4.2 Hardware Infrastructure & Convergence Progress

- Host Compute Hardware: NVIDIA GeForce RTX 4060 Laptop GPU (8GB VRAM), CUDA 12.4.
- Optimization Speed: 5.10 batches/sec (163.2 images/sec) with PyTorch Automatic Mixed Precision.
- Epoch 1: Train Loss = 0.5120 | Val Loss = 0.4124 | Val Accuracy = 81.45% | Val AUC = 0.8872
- Epoch 7 (Optimal Convergence): Train Loss = 0.0531 | Val Loss = 0.0211 | Val Accuracy = 99.29% | Val AUC = 0.9999

4.3 Empirical Test Evaluation Table (20,000 Unseen Test Images)

ompression Tie	JPEG Quality	Accuracy	AUC-ROC	Precision	Recall	F1-Score
Raw (c0)	100	99.52%	1.0000	0.9911	99.94%	0.9952
Medium JPEG (c23	23	97.94%	0.9990	0.9645	99.53%	0.9797
Heavy JPEG (c40	40	98.51%	0.9996	0.9730	99.79%	0.9853

4.4 Key Forensic Findings

1. Compression Invariance: Minimal performance delta (1.01% drop between c0 and c40).
2. Ultra-Low False Dismissal Rate: Recall remains above 99.5% across all compression conditions.
3. Discrimination Stability: AUC-ROC is >= 0.9990 on all tiers, proving robust probability calibration.

5.1 Thesis Plots & Graphical Evidence (Saved in ./figures)

- roc_curves.png / .pdf: Multi-tier ROC curves for c0, c23, c40 hugging the top-left boundary (AUC: 1.0000, 0.9990, 0.9996), visually confirming threshold independence across compression regimes.
- confusion_matrices.png / .pdf: 1x3 normalized matrices showing raw counts and percentages. True Positive rates exceed 99.5%, with False Negative rates below 0.5% under all test tiers.
- score_distributions.png / .pdf: Bimodal probability histograms showing severe polarization: real images cluster tightly at [0.00, 0.05], while fake faces cluster at [0.95, 1.00] with near-zero density at threshold 0.5.
- metrics_comparison_bar.png / .pdf: Grouped bar chart illustrating the horizontal stability of Accuracy, Precision, Recall, and F1 across c0, c23, and c40.

5.2 Full-Picture Inference Pipeline (predict.py)

The production inference engine implements an automated two-stage pipeline for real-world full-frame images:

1. Automatic Face Localization: Scans input images for face bounding boxes [x, y, w, h].
2. Contextual Expansion: Expands boxes by a 20% margin to incorporate hairline and jawline biomarkers.
3. Aspect-Ratio Preserving Fallback: If no face box is detected (e.g. tightly cropped portraits), it dynamically extracts the upper-center square to ensure facial features remain unstretched.
4. Visual Annotation: Overlays colored bounding boxes (Green = REAL, Red = FAKE) with confidence scores.

5.3 Operational Scope, Generalization & Inherent Limitations

Critical talking points for viva defense and thesis limitation sections:

- Specialized Forensic Scope: Formulated strictly as a Human Facial Forensic Detector. It is optimized to detect synthetic human facial artifacts and is intentionally not an open-domain AI illustration detector.
- GAN vs. Diffusion Artifacts: Excels on GAN upsampling artifacts (transposed convolutions). Stylized art or diffusion illustrations lack GAN grid lattice signatures and fall outside the human facial training manifold.
- Frame-Level Temporal Constraint: Evaluates single static images; does not model inter-frame optical flow.
- Upstream Dependency: In real deployment, detection accuracy relies on upstream face localization.

Defense Conclusion: The experimental results confirm that combining 2D Wavelet Packet Decomposition with spatial CNN representations via dynamic SE-attention overcomes the compression vulnerability of conventional detectors, retaining 98.51% accuracy on heavily compressed social media deepfakes.